Example

- Solve $y'' + 4y = 3\sin 2x$.
- Roots: $r = \pm 2i$; $y_c = c_1 \cos 2x + c_2 \sin 2x$.
- Since $\sin 2x$ duplicates y_c $(s = 1)$, we take $y_p = x(A\cos 2x + B\sin 2x)$ (note that initial trial is $y_t = A\cos 2x + B\sin 2x$.)
- Substituting: $y_p'' + 4y_p = -4A\sin 2x + 4B\cos 2x$.
- Matching: $-4A = 3 \Rightarrow A = -\frac{3}{4}$; $4B = 0 \Rightarrow B = 0$.

The PS is $y_p = -\dfrac{3}{4}x\cos 2x$.

GS is $\boxed{y = c_1 \cos 2x + c_2 \sin 2x - \dfrac{3}{4}x\cos 2x.}$



Example

- Solve $y'' + y' - 2y = x^2$.
- Roots: $r = -2, 1$; since 0 is not a root, $s = 0$: trial $y_p = Ax^2 + Bx + C$ (Note that $x^2 = x^2 e^{0x}$).
- Substituting $2A + (2Ax + B) - 2(Ax^2 + Bx + C) = x^2$.
- Matching: $-2A = 1,\ 2A - 2B = 0,\ 2A + B - 2C = 0$.
- Solving: $A = -\frac{1}{2},\ B = -\frac{1}{2},\ C = -\frac{3}{4}$.

The PS is $y_p = -\frac{1}{2}x^2 - \frac{1}{2}x - \frac{3}{4}$.

Gs is $\boxed{y = c_1 e^{-2x} + c_2 e^{x} - \frac{1}{2}x^2 - \frac{1}{2}x - \frac{3}{4}.}$..



Example

- Solve $y'' - y = xe^x$.
- Roots: $r = \pm 1$; since 1 is a simple root, $s = 1$: we take $y_p = x(Ax + B)e^x = (Ax^2 + Bx)e^x$. (Note that initial trial was $y_t = (Ax + B)e^x$).
- Substituting and simplifying: $y_p'' - y_p = [4Ax + (2A + 2B)]e^x$.
- Matching with xe^x: $4A = 1 \Rightarrow A = \frac{1}{4}$; $2A + 2B = 0 \Rightarrow B = -\frac{1}{4}$.

  The PS is $y_p = \dfrac{x^2 - x}{4} e^x$.

  GS is $\boxed{y = c_1 e^x + c_2 e^{-x} + \dfrac{x^2 - x}{4} e^x.}$



Example

- Solve $y'' + 2y' + 5y = e^{-x}\cos 2x$.
- Roots: $r = -1 \pm 2i$.
- $y_c = e^{-x}(c_1 \cos 2x + c_2 \sin 2x)$.
- If $y_p = e^{-x}(A\cos 2x + B\sin 2x)$, then it duplicates y_c for $c_1 = A$ and $c_2 = B$. So, take this as y_t.
- Due to $s = 1$, we need the PS as $y_p = xy_t = xe^{-x}(A\cos 2x + B\sin 2x)$.
- Substituting this in given DE, we have $A = 0$, $B = \frac{1}{4}$.

  The PS is $y_p = \frac{x}{4}e^{-x}\sin 2x$.

  GS is $\boxed{y = e^{-x}(c_1 \cos 2x + c_2 \sin 2x) + \frac{x}{4}e^{-x}\sin 2x.}$



Example

- Solve $y'' - y' - 2y = 4x - 5 + 6xe^{2x}$.
- Roots: $r = 2, -1$; $y_c = c_1 e^{2x} + c_2 e^{-x}$. Split $R = R_1 + R_2$ (superposition).
- **For $R_1 = 4x - 5$:** trial $y_{p1} = Ax + B$; solving gives $A = -2$, $B = \frac{7}{2}$.
- **For $R_2 = 6xe^{2x}$:** Since 2 is a simple root, $s = 1$, we take $y_{p2} = x(Cx + D)e^{2x}$.
- Substituting and matching gives $C = 1$, $D = -\frac{2}{3}$.

The Ps is $y_p = \left(-2x + \frac{7}{2}\right) + \left(x^2 - \frac{2}{3}x\right) e^{2x}$.

GS is $\boxed{y = c_1 e^{2x} + c_2 e^{-x} - 2x + \frac{7}{2} + \left(x^2 - \frac{2}{3}x\right) e^{2x}.}$

